

The Longest Path Problem on Permutation Graphs

Ruei-Yuan Chang¹, Cheng-Hsien Hsu², and Sheng-Lung Peng^{1*}

¹Department of Computer Science and Information Engineering
National Dong Hwa University, Hualien 97401, Taiwan

²Department of Management Information System
Takming University of Science and Technology, Taipei, 11451, Taiwan

Abstract

The longest path problem on graph G is a classic problem which is to find a simple path such that the number of vertices of the path is maximum among all possible paths of G . In this paper, we show that the longest path problem can be solved in linear time on permutation graphs.

1 Introduction

Let $G = (V, E)$ be a finite, simple, and undirected graph. Let $n = |V|$ and $m = |E|$. Let $N(v) = \{u \mid (u, v) \in E\}$. A path $P = (v_1, v_2, \dots, v_s)$ of G means $(v_i, v_{i+1}) \in E(P) \subseteq E(G)$. Let $V(P)$ be the set of vertices in P and $|V(P)| = |P|$ be the number of vertices in P . The *longest path problem* is a classical problem which is to find a path P with $|P|$ is maximum among all possible paths in graph G . There is a related problem which is called the *Hamiltonian path problem*. The Hamiltonian problem is to decide whether a graph G contains a simple path that visits each vertex of G exactly once. We may see the longest path problem is to find a Hamiltonian path in an induced subgraph of G (See Figure 1).

It is easy to see that the Hamiltonian path problem is a special case of the longest path problem. Therefore, the classes of graphs which are showed NP-complete for the Hamiltonian path problem are also NP-complete for the longest path problem. Stockmeyer *et al.* first showed that the Hamiltonian path problem is NP-complete on general graphs in 1974 [7]. There are still many NP-completeness results on some small classes of graphs. Garey *et al.* showed that the NP-completeness on planar graphs [8]. The problem on grid graphs is also NP-complete showed by Itai *et al.* [13]. In 1989, circle graphs and directed

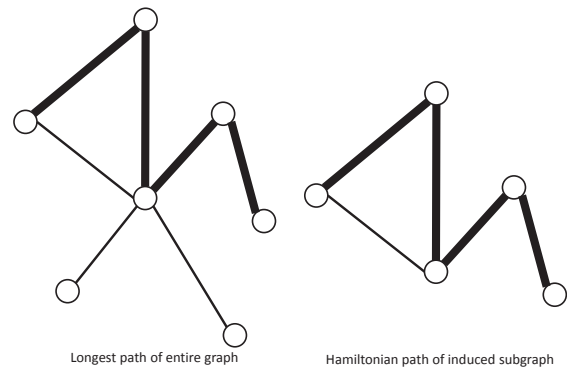


Figure 1: A graph G with longest path and a Hamiltonian path in its induced subgraph.

path graphs were proved NP-complete by Damaschke [5] and Narasimhan [18], respectively. Furthermore, Müller gives the NP-complete proofs on chordal bipartite graphs and split strongly chordal graphs [17]. Later, the NP-complete result on split graphs is given in 2004 [9]. There are some polynomial-time results. Bertossi proposed an $O(n \log n)$ algorithm for solving this problem on proper interval graphs [2]. And then, an $O(n + m)$ algorithm for this problem on interval graphs is proposed by Arikati *et al.* [1]. Damaschke *et al.* showed that the Hamiltonian path problem on co-comparability graphs can be solved in $O(n^2)$ [6]. Although the problem on grid graphs had been shown NP-complete, there are still researches that finding the necessary and sufficient conditions of the existence of Hamiltonian path on some small classes of grid graphs, namely, J -alphabet grid graphs [16], K -alphabet grid graphs [14], triangular grid graphs [10] etc.

In contrast to the Hamiltonian path problem, the longest path problem only has few polynomial-

*Corresponding author: slpeng@mail.ndhu.edu.tw

time results. The first polynomial-time algorithm for the longest path problem is on tree [3]. This algorithm was proposed by Dijkstra around 1960 but the proof was given by Bulterman *et al.*. In 2007, Uno *et al.* improved this algorithm for solving the longest path problem on weighted trees [21]. In [21], they also showed that the problem on block graphs can be solved in $O(n + m)$ time and on cacti in $O(n^2)$ time. Furthermore, interval biconvex graphs are introduced by them and the problem was solved by an $O(n^3(m + n \log n))$ algorithm. As a corollary, they showed that a longest path of a threshold graph can be found in $O(n + m)$ time. At almost the same time, Uehara *et al.* solve this problem on bipartite permutation graphs in $O(n)$ time [20]. Later, an $O(n^5)$ time algorithm for ptolomaic graphs is proposed by Takahara *et al.* [19]. Recently, Ioannidou *et al.* proposed an $O(n^4)$ algorithm for this problem on interval graphs [12]. Their algorithm is based on dynamic programming technique. For a larger class of graphs, cocomparability graphs, Mertziou *et al.* solved the problem in $O(n^4)$ time. More recent, Keshavarz-Kohjerdi *et al.* proposed a linear-time algorithm for rectangular grid graphs [15]. For digraphs, there is an $O(n^3)$ -time algorithm for the problem on complete m -partite digraphs proposed by Gutin [11].

In this paper, we study the longest path problem on permutation graphs. We propose a linear-time algorithm for the problem and the idea is presented in the next section. The conclusion is given in the last section.

2 Permutation Graphs

Let π be a permutation of $(1, \dots, n)$. The *matching diagram* of π is obtained as follows. Write the integers $(1, \dots, n)$, horizontally from left to right. Underneath, write the integers $(\pi_1, \dots, \pi_n)$, also horizontally from left to right. Draw n straight line segments connecting the two 1's, the two 2's, and so on. A graph is a *permutation graph* if it is isomorphic to the intersection graph of the line segments of a matching diagram. Figure 2 shows an example of a permutation graph with a matching diagram.

A shortest path between two vertices of a graph G is a path with fewest vertices among all possible paths between the two vertices. Let D (respectively, H) be the diameter (respectively, Hamiltonian path) of G . Then we can easily obtain the following lemma.

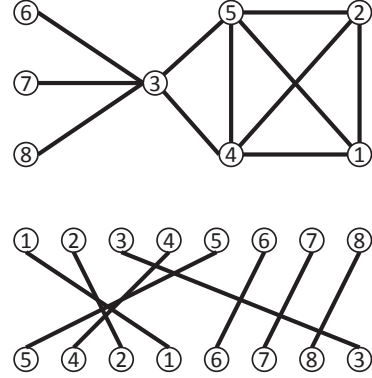


Figure 2: A permutation graph with a matching diagram.

Lemma 1. *Let G be a permutation graph and P be a longest path of G . Then $|D| \leq |P| \leq |H| = |V|$.*

In Figure 2, it is not hard to see that the diameter D is $(1, 5, 3, 8)$. The diameter can be obtained by scanning π . We check the four shortest paths: v_1 to v_n , v_1 to v_{π_n} , v_{π_1} to v_n , and v_{π_1} to v_{π_n} . Let $P_{1,n}$, P_{1,π_n} , $P_{\pi_1,n}$, and P_{π_1,π_n} denote the four shortest paths. Due to the property of permutation graphs, the vertices in $V \setminus D$ are all adjacent to D . The idea of obtaining diameter is to scan π from left to right by connecting the furthest vertex. This gives us an idea as follows. We scan π from left to right by connecting the “nearest” vertex and jumping back when there is no neighbor. We keep scanning π until to the last vertex. Is the path found by the above process a longest path of G . In other words, we are interesting at that whether the path obtained by doing modified DFS on π is a longest path of G ?

Lemma 2. *Let G be a permutation graph and π be the permutation. Then one of $P_{1,n}$, P_{1,π_n} is longer than the other one.*

Proof. Note that the four shortest paths $P_{1,n}$, P_{1,π_n} , $P_{\pi_1,n}$, and P_{π_1,π_n} are obtained by connecting the furthest vertex. It is obviously that v_n is adjacent to v_{π_n} . Considering $P_{1,n}$, suppose v_x is the previous vertex of v_n in $P_{1,n}$. Since the algorithm for finding the shortest path in permutation graphs is always connecting the furthest vertex, $P_{1,n}$ and P_{1,π_n} both contain the shortest path from v_1 to v_x . If $v_{\pi_n} \in P_{1,n}$, then $P_{1,\pi_n} \subseteq P_{1,n}$. Otherwise, v_x is not adjacent to v_{π_n} . Thus, P_{1,π_n} is

equal to adding v_{π_n} behind $P_{1,n}$. Hence we complete the proof. $\square$

By similar argument, we can obtain the following lemma.

Lemma 3. *Let G be a permutation graph and π be the permutation. Then one of $P_{\pi_1,n}$, and P_{π_1,π_n} is longer than the other one.*

Let D be the diameter found in the four shortest paths. We try to find the longest path by scanning π from left to right. At beginning, we check the start vertex of D . Without loss of generality, we assume the vertex is v_1 . Then we add v_1 to P (initially, $P = \phi$) and put v_1 into a stack S . Next, we choose the minimum vertex v_{π_1} on the other side to connect. At the same time, we add it to P and S . Next we connect v_{π_1} to $\min_x \{v_x \mid (v_x, v_{\pi_1}) \in E(G)\}$ where v_x is unvisited. Continue this operation until $\min_s \{v_s \mid (v_s, v_t) \in E(G)\}$ is empty where v_t is the tail vertex of path P_1 and v_s is not visited. Then we pop the stack and check whether the vertex still has unvisited neighbors. If it did, then we keep scanning π . If not, then we keep popping the stack. Finally, we stop scanning π and the longest path is obtained. In top idea, we are doing extension on the diameter. We change the connecting scheme from “furthest” to “nearest” vertex for adding the vertices in $V \setminus D$ to extend D as longer as possible. If all vertices in $V \setminus D$ are added, then we obtain the Hamiltonian path H . If the length of the longer one of $P_{1,n}$, P_{1,π_n} is equal to the longer one of $P_{\pi_1,n}$, P_{π_1,π_n} , then we will do the modified DFS starting from v_{π_1} and check which one of the two paths is longer. Summarizing the above operations, we obtain our algorithm, LongestPath.

It is obvious that the time complexity of LongestPath is $O(n+m)$. By the discussion above, we have the following theorem.

Theorem 1. *Algorithm LongestPath finds a longest path P of a permutation graph G in $O(n+m)$ time.*

3 Conclusion

In this paper, we present an $O(n+m)$ algorithm to solve the longest path problem on permutation graphs. By using the permutation diagram, we obtain the longest path efficiently. It is interesting whether this approach can be extended to solve the same problem on AT-free graphs.

Algorithm ModifiedDFS

Data: the first vertex v of P

Result: A path P .

Stack $S = \phi$;

$P = \phi$;

forall $v_i \in V$ **do**

$\lfloor$ $cover[v_i] = 0$;

if $v = v_1$ **then**

$\lfloor$ $line = 1$;

$\lfloor$ $v_{now} = v_1$;

else

$\lfloor$ $line = 0$;

$\lfloor$ $v_{now} = v_{\pi_1}$;

$cover[v_{now}] = 1$;

add v_{now} to S ;

add v_{now} to P ;

while $cover[n] = 0$ or $cover[\pi^{-1}(\pi_n)] = 0$

do

if $line = 0$ **then**

$\lfloor$ $v_{next} = \min_i \{v_i \mid cover[i] = 0 \text{ and } (v_i, v_{now}) \in E(G)\}$;

else

$\lfloor$ $v_{next} = \min_i \{v_{\pi_i} \mid cover[i] = 0 \text{ and } (v_{\pi_i}, v_{now}) \in E(G)\}$;

if $v_{next} = \phi$ **then**

 pop S ;

 remove v_{now} from P ;

else

$\lfloor$ $cover[v_{next}] = 1$;

 push v_{next} to S ;

 add v_{next} to P ;

$\lfloor$ $line = (line + 1) \bmod 2$;

return P ;

Algorithm LongestPath

Data: the permutation π of G .

Result: A longest path P of G .

$P = \phi$;

Find $P_{1,n}$, P_{1,π_n} , $P_{\pi_1,n}$, and P_{π_1,π_n} ;

$P_1^* = \max_{length}(P_{1,n}, P_{1,\pi_n})$;

$P_{\pi_1}^* = \max_{length}(P_{\pi_1,n}, P_{\pi_1,\pi_n})$;

if $|P_1^*| \neq |P_{\pi_1}^*|$ **then**

 Let v be the first vertex of the longer one of P_1^* and $P_{\pi_1}^*$;

$P = ModifiedDFS(v)$;

else

$P_1 = ModifiedDFS(v_1)$;

$P_2 = ModifiedDFS(v_{\pi_1})$;

$P = \max_{length}(P_1, P_2)$;

return P ;

References

- [1] S.R. Arikati, C.P. Rangan: Linear algorithm for optimal path cover problem on interval graphs. *Information Processing Letters*, **35(3)**, 149-153, 1990.
- [2] A A. Bertossi: Finding Hamiltonian circuits in proper interval graphs. *Information Processing Letters*, **17(2)**, 97-101, 1983.
- [3] R. W. Bulterman, F. W. van der Sommen, G. Zwaan, T. Verhoe, A. J. M. van Gasteren, W. H. J. Feijen: On computing a longest path in a tree. *Information Processing Letters*, **81(2)**, 93-96, 2002.
- [4] R.-Y. Chang, S.-L. Peng: Minimum edge ranking spanning tree problem on interval graphs, *Workshop on Combinatorial Mathematics and Computation Theory 2011*, 200-203, 2011.
- [5] P. Damaschke: The Hamiltonian circuit problem for circle graphs is NP-complete. *Information Processing Letters*, **32(1)**, 1-2, 1989.
- [6] P. Damaschke, J.S. Deogun, D. Kratsch, G. Steiner: Finding Hamiltonian paths in cocomparability graphs using the bump number algorithm. *Order*, **8**, 383-391, 1992.
- [7] M.R. Garey, D.S. Johnson, L. Stockmeyer: Some simplified NP-complete problems. *Proceedings of the sixth annual ACM symposium on Theory of computing*, 47-63, 1974.
- [8] M.R. Garey, D.S. Johnson, R.E. Tarjan: The planar Hamiltonian circuit problem is NP-complete. *SIAM Journal on Computing*, **5**, 704-714, 1976.
- [9] M.C. Golumbic: Algorithmic graph theory and perfect graphs (Annals of Discrete Mathematics, Vol. 57). North-Holland Publishing Co., 2 edition, 2004.
- [10] V.S. Gordon, Y.L. Orlovich, F. Werner: Hamiltonian properties of triangular grid graphs, *Discrete Mathematics*, **308**, 6166-6188, 2008.
- [11] G. Gutin: Finding a longest path in a complete multipartite digraph, *SIAM J. Discrete Mathematics*, **6(2)**, 270-273, 1993.
- [12] K. Ioannidou, G.B. Mertzios, S. Nikolopoulos: The longest path problem is polynomial on interval graphs, in: Proc. of 34th Int. Symp. on Mathematical Foundations of Computer Science, 5734, Springer-Verlag, Novy Smokovec, High Tatras, Slovakia, 403-414, 2009.
- [13] A. Itai, C. H. Papadimitriou, J. L. Szwarcwarter: Hamiltonian paths in grid graphs. *SIAM Journal on Computing*, **11**, 676-686, 1982.
- [14] F. Keshavarz-Kohjerdi, A. Bagheri: Hamiltonian paths in K -alphabet grid graphs, *Journal of Computing*, **3(9)**, ISSN 2151-9617, September 2011,.
- [15] F. Keshavarz-Kohjerdi, A. Bagheri: A. Asgharian-Sardroud, A linear-time algorithm for the longest path problem in rectangular grid graphs, *Discrete Applied Mathematics*, **160(3)**, 210-217, 2012.
- [16] F. Keshavarz-Kohjerdi, A. Bagheri, B. Tayfe-Rezaie: Hamiltonian paths in J -alphabet grid graphs, Proc. 2ndWorld Conference on Information Technology, to appear, 2011.
- [17] H. Müller: Hamiltonian circuits in chordal bipartite graphs, *Discrete Mathematics*, **156(1-3)**, 291-298, 1996.
- [18] G. Narasimhan: A note on the Hamiltonian circuit problem on directed path graphs, *Information Processing Letters*, **32(4)**, 167-170, 1989.
- [19] Y. Takahara, S. Teramoto, R. Uehara: Longest path problems on Ptolemaic graphs. *IEICE - Transactions on Information and Systems*, **E91-D(2)**, 170-177, 2008.
- [20] R. Uehara, G. Valiente: Linear structure of bipartite permutation graphs and the longest path problem, *Information Processing Letters*, **103(2)**, 71-77, 2007.
- [21] R. Uehara, Y. Uno: On computing longest paths in small graph classes, *Internat. J. Found. Comput. Sci.*, **18(5)**, 911-930, 2007.